GPAS: FIXED-OBJECTIVE, OPTIMIZER-AWARE TASK SAMPLING FOR MULTI-TASK LLM POST-TRAINING

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ABSTRACT

Multi-task LLM post-training uses a task mixture for two jobs. Objective weights say how much each teacher loss counts, while sampling probabilities decide where rollout and teacher compute go. An adaptive mixture changes both at once, which makes compute efficiency and target choice hard to separate.

We study task-homogeneous updates that draw one task per step. The sampled gradient is multiplied by its objective weight divided by its sampling probability, the standard unbiased estimator for a fixed weighted objective. Standard AdamW squares the same multiplier in its second-moment update, so a new proposal also changes the preconditioner's population target. A one-line variant, moment-consistent AdamW, uses the multiplier once. At a frozen training state, its second-moment target depends on the objective weights and task gradients and stays independent of the proposal. Gradient-Preconditioned Adaptive Sampling (GPAS) then samples each task in proportion to its objective weight and preconditioned gradient scale. Cost-GPAS adds an inverse square-root correction for measured step time. Both methods reuse the gradient and timing from the current step and keep at most two scalars per task.

We pair these rules with a matched four-teacher MOPD protocol that holds the objective weights fixed and compares teacher loss per token and GPU hour. A separate curriculum pairing replays the same objective-weight path so that only allocation changes. Controlled tests and frozen-gradient replay measure moment drift, estimator error, scale staleness, importance weights, and clipping. These measurements test when the local allocation rule predicts end-to-end MOPD efficiency.

1 INTRODUCTION

Post-training one student on several capabilities means choosing which teacher or reward service to call at every step. A task mixture usually controls two decisions at once: how much each task counts in the training objective and how often the system spends rollout, teacher, and optimizer time on that task. Once the mixture follows loss gaps, gradient norms, or a curriculum, the training target moves with the compute schedule.

We separate these decisions. Objective weights λ define the weighted teacher-loss objective

$$F(\theta) = \sum_i \lambda_i L_i(\theta). \tag{1}$$

Sampling probabilities q decide which task receives the next expensive step. This distinction also keeps capability evaluation conceptually separate: one teacher loss can affect several downstream domains (Li et al., 2026). An experiment can now change allocation while keeping the teacher-loss objective fixed.

Our execution regime selects one task before generation and forms a task-homogeneous batch with a fixed valid-token budget. Equal token budgets still leave two differences that matter for allocation: the AdamW-scaled gradient produced by a task and the wall-clock time required for its step. Our warm start measures both quantities before adaptive sampling begins. The main study reports their taskwise range instead of using prompt or token share as a proxy for optimization compute. This regime matches a worker that keeps one teacher in memory or calls one task-specific reward service.

For task i, the factor λ_i/q_i gives the standard unbiased gradient estimate. AdamW squares this factor when it updates its second moment, which makes the preconditioner depend on the proposal. Lahire (Lahire, 2023) identified the broader issue: the gradient-norm proposal that is optimal for SGD loses that property under an adaptive optimizer. We give a constructive AdamW solution.

Moment-consistent AdamW uses $(\lambda_i/q_i)G_i^2$ as its second-moment observation. At a frozen training state, the population target of this update is independent of q. GPAS measures each task's preconditioned gradient scale s_i in the lagged AdamW geometry and selects $q_i \propto \lambda_i s_i$. If task i costs c_i seconds, Cost-GPAS selects $q_i \propto \lambda_i s_i/\sqrt{c_i}$. The second rule minimizes a simple cost–gradient product and is the one-draw form of cost-constrained Neyman allocation.

The evaluation mirrors the method. Fixed-λ runs compare Uniform, raw-gradient importance sampling, GPAS, Cost-GPAS, and a corrected D^3 proposal under one teacher-loss objective. A paired curriculum experiment replays the same λ sequence with two allocation rules. A standard AdamW arm measures the role of the moment update. Teacher-loss curves provide the optimization endpoint, while capability benchmarks record the final model profile.

Our contributions are:

- A fixed-objective protocol for one-task updates: standard importance correction fixes the weighted gradient target, and a one-line AdamW change gives the second moment a proposal-independent population target.
- GPAS and Cost-GPAS, which allocate tasks using preconditioned gradient scale and measured runtime. Each method reuses quantities already produced by training and keeps at most two scalars per task.
- A matched MOPD study that separates target schedules from compute allocation, including a replayed D^3 objective-weight path and a standard-AdamW ablation.
- Controlled and frozen-gradient tests that connect each local claim to moment drift, estimator error, staleness, clipping, and GPU-hour efficiency.

2 Related Work

Objective weights and multi-task optimization. Multi-task methods often tune loss weights or edit gradient directions to change the capability trade-off (Sener & Koltun, 2019; Yu et al., 2020; Liu et al., 2024; Chen et al., 2018). Large empirical comparisons found carefully tuned scalarization competitive with specialized multi-task optimizers in their evaluated settings (Kurin et al., 2022; Xin et al., 2022). These results support our decision to state the weighted objective explicitly and evaluate allocation under fixed weights. GPAS also accepts a time-varying objective supplied by a curriculum; the proposal is computed after each weight update.

Adaptive data and prompt schedules for language models. Online Data Mixing uses a bandit to update pretraining-domain proportions (Albalak et al., 2023). Recent post-training methods schedule distributions from average advantage, skip low-value rollouts, or select prompts near the current learning frontier (Wang et al., 2025; Zheng et al., 2025; Gao et al., 2025). These methods optimize a domain or prompt policy with signals tied to learning progress. Our question concerns task-level teacher compute: given stated objective weights, which proposal gives an efficient unbiased gradient in the current optimizer geometry? The fixed-target comparisons place heuristic schedules and GPAS under the same objective.

Multi-teacher post-training. MOPD routes student rollouts to task teachers and trains a shared student with dense token feedback (Ma et al., 2026). Open-MOPD identifies token count, reward scale, and stale rewards as sources of unequal update budgets (Gao et al., 2026). D^3-MOPD uses teacher-loss gap and descent velocity to update the domain mixture (Sun et al., 2026). Teacher mixtures can also create cross-domain capability transfer and a mixture-dependent seesaw (Li et al., 2026). The cross-domain transfer results motivate our use of teacher loss for objective comparisons and separate benchmark reporting for capability. We compare D^3 in two roles: as a proposal under a fixed objective and as a curriculum whose recorded objective-weight path is replayed with GPAS.

Adaptive importance sampling. Gradient-norm importance sampling reduces the second moment of an SGD estimate (Alain et al., 2015; Katharopoulos & Fleuret, 2018). Cost-aware sampling adds a square-root cost correction and provides convergence guarantees beyond convex objectives (Mohri et al., 2026). The same allocation shape appears in classical Neyman allocation (Neyman, 1934; Cochran, 1977). Online importance samplers face partial feedback because only the selected item receives a fresh scale observation. Bandit formulations provide exploration and regret controls for this setting (Namkoong et al., 2017; Salehi et al., 2017; Borsos et al., 2018). Our probability floor plays the practical exploration role and also bounds the importance multiplier; scale age and full-task refreshes measure the remaining feedback error.

Sampling with adaptive optimizers. Lahire showed that the gradient-norm proposal for SGD loses its optimality for RMSProp and Adam and argued that sampling should match the optimizer (Lahire, 2023). GPAS gives a constructive answer for AdamW: a proposal-independent second-moment target followed by an allocation rule in the resulting preconditioned geometry. The short-memory intuition is related to Adam's sign and scale behavior described by Balles & Hennig (2018).

3 A Fixed Objective under Task Sampling

There are M tasks. Task i has loss L_i and objective weight $\lambda_i \geq 0$, with $\sum_i \lambda_i = 1$. These weights define Equation 1. Each step draws one task $I \sim q$, forms a task-homogeneous batch, and computes a task gradient G_I. All expectations in this section condition on the model, optimizer state, and sampling information available before the draw.

Our MOPD batches use a fixed valid-token budget. Tokens are averaged inside a response, followed by an average over responses. This gives one consistent definition of G_i even when response lengths differ. The analysis also applies to other within-task aggregation rules once they are fixed in advance.

For every task with positive objective weight, we require $q_i > 0$ and set

$$w_i = \frac{\lambda_i}{q_i}, \qquad \widehat{g} = w_I G_I. \tag{2}$$

Lemma 1 (Standard importance correction)**.** *The corrected gradient has mean*

$$\mathbb{E}[\widehat{g}] = \sum_i \lambda_i \mathbb{E}[G_i] = \nabla F(\theta). \tag{3}$$

This familiar importance-sampling result fixes the weighted gradient target for any proposal. A curriculum may change λ between steps; the same statement applies after each change.

The correction interacts with AdamW through its second moment. We use $w_I G_I$ for the first-moment observation and $w_I G_I^2$ for the second-moment observation:

$$\begin{aligned} m^+ &= \beta_1 m + (1-\beta_1) w_I G_I, \\ v^+ &= \beta_2 v + (1-\beta_2) w_I G_I^2, \end{aligned} \tag{4}$$

where the square is element-wise. Bias correction, weight decay, and the parameter update stay unchanged. We call this rule moment-consistent AdamW.

Proposition 1 (Proposal-independent second-moment target)**.** *At a fixed training state, let $S_i = \mathbb{E}[G_i^2]$. The mean observation in the moment-consistent second-moment update is*

$$\sum_i \lambda_i S_i, \tag{5}$$

which is independent of q. Standard AdamW observes $(w_I G_I)^2$ and has mean

$$\sum_i \frac{\lambda_i^2}{q_i} S_i. \tag{6}$$

If the task-gradient distributions and weights were held fixed, these two means would be the respective population targets of the exponential average.

Appendix B.1 gives the two-line calculation. The claim concerns the second-moment target at a fixed state. Task order can still produce different parameter paths, and the target moves as the model learns.

The short-memory limit gives useful intuition. Ignore momentum, weight decay, and ϵ, and let the current observation dominate v. For a positive importance multiplier, the two scaled updates become

$$\frac{wG}{\sqrt{w^2G^2}} = \operatorname{sign}(G), \qquad \frac{wG}{\sqrt{wG^2}} = \sqrt{w}\,\operatorname{sign}(G). \tag{7}$$

Standard AdamW removes the multiplier in this idealized limit, while the moment-consistent rule retains its square root. With long second-moment memory, the moment-consistent denominator tracks Equation 5; the standard denominator moves with q through Equation 6. This sign-and-scale view follows the Adam interpretation of Balles & Hennig (2018).

4 Optimizer-Aware Task Allocation

Once the second-moment target is independent of the proposal, task allocation can be compared in one fixed AdamW geometry. Let $\bar{v}$ be the bias-corrected second moment before the task draw. We define the preconditioned gradient scale of task i as

$$s_i^2 = \mathbb{E}\left\| \frac{G_i}{\sqrt{\bar{v}} + \epsilon} \right\|_2^2. \tag{8}$$

This is the squared gradient norm after AdamW's coordinate scaling and before the importance multiplier.

For one corrected task draw, the proposal-dependent part of the scaled gradient's second moment is

$$V(q) = \sum_i \frac{\lambda_i^2 s_i^2}{q_i}. \tag{9}$$

Its minimum occurs at

$$q_i^{\mathrm{GPAS}} = \frac{\lambda_i s_i}{\sum_j \lambda_j s_j}. \tag{10}$$

We call this rule Gradient-Preconditioned Adaptive Sampling (GPAS). Under the identity scale it reduces to gradient-norm importance sampling. The derivation and its centered-error form appear in Appendix B.2.

This local quantity connects directly to optimization progress. For a smooth loss and a fixed diagonal preconditioner, the usual one-step descent bound contains $V(q)$ as its only proposal-dependent term. The bound uses the uncentered second moment, which explains the definition in Equation 9. Frozen-gradient replay tests whether the proposal ranking from $V(q)$ predicts short-horizon teacher-loss decrease.

Task time changes the useful allocation. Let c_i be the measured duration of a task-i step. The expected time per draw is $\sum_i q_i c_i$, so a GPU budget B permits about $B/(\sum_i q_i c_i)$ steps. Substituting this count into the standard stochastic-rate bound leaves the proposal-dependent product

$$J(q) = \left(\sum_i q_i c_i\right) V(q). \tag{11}$$

Minimizing this compute-efficiency proxy gives

$$q_i^{\text{Cost-GPAS}} \propto \frac{\lambda_i s_i}{\sqrt{c_i}}. \tag{12}$$

This is cost-constrained Neyman allocation with tasks as strata (Neyman, 1934; Cochran, 1977; Mohri et al., 2026). Appendix B.3 states the smoothness argument and the closed-form calculation. GPU-hour curves test the proxy against end-to-end training.

The scale contains a mean-update term and within-task gradient noise. A large token batch may make the mean term dominant, in which case GPAS spends more steps on tasks that currently move the student farther in AdamW coordinates. Correlated tokens and variable response counts can keep the noise term large. Our gradient-bank experiment estimates both terms with microbatch resampling instead of assuming either regime.

Table 1: The comparison separates the objective, proposal signal, cost term, and AdamW second-moment rule. "MC" denotes moment-consistent AdamW.

Method	Objective weights	Proposal signal	Step time	Second moment
Uniform	fixed λ	uniform	—	MC
Gradient-Norm IS	fixed λ	raw gradient scale	—	MC
GPAS	fixed λ	preconditioned scale	—	MC
Cost-GPAS	fixed λ	preconditioned scale	measured	MC
D^3-Proposal-IC	fixed λ	loss gap and velocity	—	MC
Native D^3	scheduled λ	loss gap and velocity	—	native
D^3 target + GPAS	replayed λ	preconditioned scale	—	MC
GPAS + standard AdamW	fixed λ	preconditioned scale	—	squared weight

Run a round-robin warm start to initialize one scale estimate and one optional time estimate per task. At each step: (1) form the raw proposal from Equation 10 or 12; (2) apply Equation 14 and draw one task; (3) run rollout, teacher evaluation, and backpropagation for that task; (4) compute $w = \lambda_I / q_I$ and take a moment-consistent AdamW step; (5) update the selected task's scale and time averages. Save these averages with the optimizer state.

Algorithm 1: Online GPAS / Cost-GPAS. GPAS adds task selection and scalar tracking to an existing task-homogeneous MOPD loop. The current backward pass supplies every statistic used by the next proposal.

Online estimates and exploration. After a task-i step, GPAS updates a moving average of the observed squared scale,

$$\widehat{s}_i^2 \leftarrow \gamma \widehat{s}_i^2 + (1-\gamma) \left\| \frac{G_i}{\sqrt{\widehat{v}} + \epsilon} \right\|_2^2. \tag{13}$$

Cost-GPAS maintains the same type of average for step time. Because only the selected task receives a fresh observation, stale low estimates can reinforce low sampling probabilities. A round-robin warm start and an explicit floor provide exploration. If p is the raw GPAS proposal, we use

$$q_i = (1 - M q_{\min}) p_i + q_{\min}, \qquad 0 < q_{\min} < 1/M. \tag{14}$$

This rule gives $q_i \geq q_{\min}$ and $w_i \leq \max_i \lambda_i / q_{\min}$. For four uniform objective weights and $q_{\min} = 0.05$, the largest possible multiplier is 5. The experiments sweep the floor and report scale age, multiplier distributions, update norms, and clip rates.

Drawing several independent tasks in one step gives the same proposal for each draw; their corrected gradients are averaged before the optimizer update.

5 Experiments

The experiments test one chain of claims. The warm start first measures whether tasks differ in preconditioned gradient scale and end-to-end step time. Frozen-gradient replay then checks the local moment and allocation predictions at real LLM states. Matched MOPD runs test whether those predictions translate to teacher-loss progress per token and GPU hour. Controlled problems serve as implementation checks and stress tests for the underlying formulas.

5.1 Four-teacher MOPD setup

The student is a fixed revision of Qwen3-1.7B (Qwen Team, 2025). Four same-origin teachers specialize in mathematics, code, instruction following, and science question answering. The run manifest records model and tokenizer revisions, teacher checkpoints, data revisions, optimizer settings, hardware, and seeds. Each teacher is also evaluated on its own domain before distillation.

Every step draws one task and builds a task-homogeneous batch with 65,536 valid response tokens. Tokens are averaged inside a response, followed by an average over responses. The logs retain response, prompt, and valid-token counts, which allows aggregation choices to be audited. The step timer covers rollout, teacher evaluation, backpropagation, and the optimizer update.

Table 2: Matched MOPD runs. The curriculum pair reuses the same realized objective-weight path.

Method	Objective λ	Proposal q	Seeds
Uniform	$1/4$	$1/4$	3
Gradient-Norm IS	$1/4$	$\propto \lambda\widehat{r}$	3
GPAS	$1/4$	$\propto \lambda\widehat{s}$	3
Cost-GPAS	$1/4$	$\propto \lambda\widehat{s}/\sqrt{\widehat{c}}$	3
D^3-Proposal-IC	$1/4$	D^3 signal + correction	3
Native D^3	q_t	D^3 signal	2
D^3 target + GPAS	replayed λ_t	$\propto \lambda_t\widehat{s}$	2
GPAS + standard AdamW	$1/4$	$\propto \lambda\widehat{s}$	2

Training uses an 8,192-token response cap, temperature 1, top-p 1, an eight-step round-robin warm start, $\gamma = 0.95$, and $q_{\min} = 0.05$. The shared emergency norm clip is applied to G_i before importance correction. The optimizer receives the clipped task gradient and w separately, uses wG_i for its first moment, and uses wG_i^2 for its second moment. Clip events measure divergence from the original teacher-loss gradient field; the corrected estimate still targets the corresponding clipped update field.

A 300-step pilot pairs Uniform and GPAS under one seed. The main study runs 400 steps and reuses the pilot checkpoints. The five fixed-objective methods use three paired seeds, and the moment comparison uses two. Runs within a seed share initialization, per-task data order, and evaluation prompts. Held-out teacher loss is evaluated every 50 steps, with capability benchmarks at the endpoint.

5.2 MATCHED METHODS

Table 2 separates fixed-objective allocation from a curriculum. Rows through D^3-Proposal-IC use $\lambda_i = 1/4$ and moment-consistent AdamW. Gradient-Norm IS uses raw gradient scale. D^3-Proposal-IC converts the D^3 loss-gap and descent-velocity signal into q, followed by the same λ_i/q_i correction as every fixed-objective method.

Native D^3 uses its schedule as both objective weights and sampling probabilities. For each seed, the paired D^3 target + GPAS run replays the recorded sequence $\lambda_1, \ldots, \lambda_T$ and changes only the task draws. This pairing holds the realized objective-weight path fixed. The final row keeps the GPAS proposal and first-moment correction and returns the second moment to $w^2G_i^2$.

5.3 END-TO-END ENDPOINTS

The primary endpoint is weighted held-out teacher loss

$$L(N) = \sum_i \lambda_i \ell_i(N), \tag{15}$$

plotted against cumulative valid response tokens and measured GPU hours. We report normalized loss AUC on the common resource interval, final weighted loss, every per-task loss, and GPU hours needed to reach the final Uniform loss. Mean curves and standard errors use paired seeds.

The fixed-objective rows give direct mechanism comparisons. Gradient-Norm IS versus GPAS tests the optimizer metric. GPAS versus Cost-GPAS tests the runtime correction through token-efficiency and time-efficiency curves. GPAS versus its standard-AdamW arm tests the second-moment rule. The replayed D^3 pair tests allocation under one curriculum path.

Capability reporting includes MATH-500 greedy pass@1, a frozen post-cutoff LiveCodeBench pass@1 slice, IFBench strict accuracy, and GPQA-Diamond average@4. Per-task teacher losses and capability scores remain primary for the native curriculum rows because their objective changes during training.

5.4 MECHANISM, FEEDBACK, AND STABILITY

The Uniform run supplies checkpoints after the round-robin warm start, near the middle, and near the end of training. At each checkpoint we collect 16 task batches per task. Eight batches estimate each proposal and eight evaluate it; swapping the folds and averaging removes the same-bank advantage. Rollouts and teacher log probabilities are cached so the accounting includes their collection cost.

The gradient bank answers four practical questions. Raw and preconditioned gradient norms are compared with the exact task-dependent AdamW update, defined at a fixed state as $\Delta\theta(G_i)-\Delta\theta(0)$. Correlations are reported both within each task and after pooling tasks. The held-out fold evaluates $V(q)$ and $J(q)$ for every proposal. Sampled-coordinate replay compares the population second-moment targets and finite-memory state drift under standard and moment-consistent AdamW. Fixed random projections provide pairwise cosine similarities between task mean gradients.

Microbatch resampling separates the squared preconditioned mean gradient from within-task noise, which tests whether the 65,536-token regime is mean-driven. Short branches from each checkpoint run the candidate proposals for the same small step budget and compare predicted $V(q)$ and $J(q)$ rankings with actual held-out loss decrease. This supplies the empirical bridge from a local second-moment criterion to training progress.

An offline sampler replay sweeps $\gamma \in \{0.90, 0.95, 0.99\}$ and $q_{\min} \in \{0.01, 0.025, 0.05, 0.10\}$. It reports scale age, full-refresh scale error, the contribution of the floor to $J(q)$, and the frequency of extreme importance multipliers. End-to-end runs report per-task warm-start ranges for s_i and c_i, multiplier and corrected-update quantiles, clip rate, numerical overflow, and runtime-estimation error. A small optimizer microbenchmark also records step latency and peak memory for standard and moment-consistent AdamW.

5.5 Controlled checks

The first controlled problem uses three zero-mean Gaussian task gradients with deliberately misaligned raw and AdamW coordinate scales. Every proposal targets $\lambda_i = 1/3$. The estimator calculation and 20,000 Monte Carlo trials use 16 corrected task draws per trial. This construction is a worst-case illustration for raw-gradient sampling; its purpose is to expose the metric mismatch rather than model a typical LLM gradient.

In this construction, the calculated and simulated AdamW-scaled error ratios are 0.717 for GPAS and 6.861 versus 6.868 for Gradient-Norm IS, relative to Uniform. Cost-GPAS reaches a calculated $J(q)$ ratio of 0.893 and a simulated ratio of 0.889. The close calculation–simulation agreement checks the estimator implementation; the deliberately large raw-gradient ratio comes from the chosen coordinate mismatch.

The final panel draws one million task gradients from a proposal different from λ. Relative second-moment bias is

$$\frac{\|\mathbb{E}[O_q] - \mathbb{E}[O_{q=\lambda}]\|_2}{\|\mathbb{E}[O_{q=\lambda}]\|_2}, \tag{16}$$

where $O_q = w^2 G_I^2$ for standard AdamW and $O_q = w G_I^2$ for the moment-consistent rule. Agreement between calculation and simulation is an implementation check for Proposition 1. With one million draws, standard AdamW has second-moment bias 0.816, close to the calculated 0.818, while moment-consistent AdamW has bias 0.002. Both first-moment estimates have bias 0.008 under the same draws.

To measure sensitivity beyond one construction, a separate stress test draws 10,000 four-task, 16-coordinate geometries from a fixed broad log-uniform distribution. It reports the full error-ratio distribution, quantiles, and the fraction of cases where raw-gradient sampling exceeds Uniform. The scan is a robustness check over synthetic geometries; MOPD gradients determine external validity.

Raw-gradient sampling exceeds Uniform in 28.2% of the sampled geometries, with median ratio 0.971 and 95th percentile 1.065. GPAS attains the closed-form minimum in every geometry; its median ratio is 0.895 and its 95th percentile is 0.982.

6 Limitations and Conclusion

The theory fixes the model and optimizer state before comparing proposals. In training, task order changes future gradients, so two unbiased samplers can follow different parameter paths. The

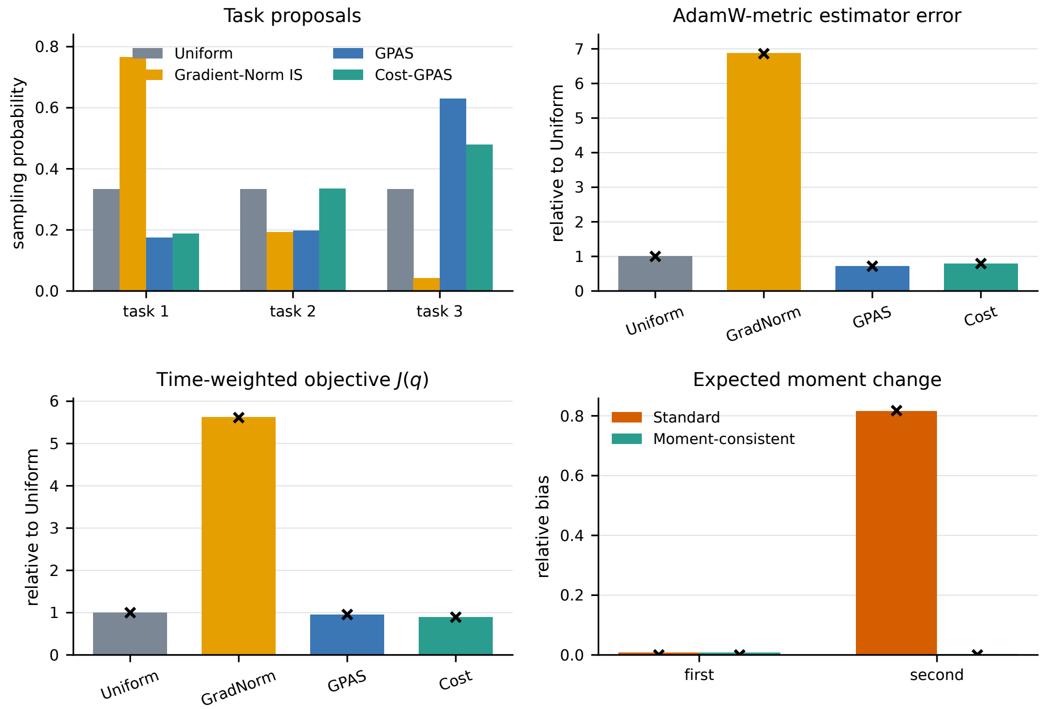

Figure 1: A constructed three-task check. The first three panels compare proposals, AdamW-scaled estimator error, and $J(q)$. Black markers show calculated values. The last panel reports the relative bias from Equation 16. Lower values are better in the last three panels.

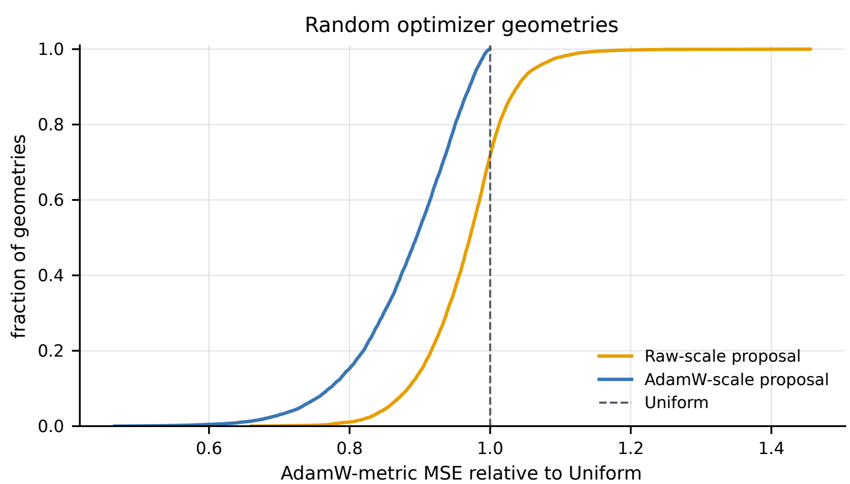

Figure 2: Random-geometry stress test. Each point compares a proposal with Uniform in the AdamW-scaled second-moment criterion. The fixed generation distribution, seed, and all 10,000 outcomes are released with the figure.

population-target statement in Proposition 1 gives a common local geometry rather than a shared trajectory.

Online scale estimates receive partial feedback. A low estimate can grow stale while its task is sampled less often. The round-robin warm start, proposal floor, age logs, and periodic full refreshes make this loop visible, while the floor trades local allocation quality for exploration and bounded importance multipliers. More formal online guarantees would require assumptions about how quickly task gradients change during post-training.

GPAS allocates from the uncentered preconditioned second moment. In a mean-dominated large batch, this favors tasks that currently make a large AdamW-scaled update. Strong within-task noise changes that interpretation, so the gradient bank reports the mean and noise terms separately. Cost-GPAS adds another approximation: $J(q)$ is a smooth-optimization proxy built from a fixed preconditioner and average step times. Teacher-loss curves against GPU hours provide the deciding empirical test.

Gradient clipping also changes the object being estimated. Importance correction remains exact for the clipped update field, while clip events move that field away from the original teacher-loss gradient. Per-task clip rates, update norms, and multiplier distributions define the practical scope of the fixed-objective interpretation.

The central design is simple: state the teacher-loss objective, then allocate expensive task steps for that objective. Moment-consistent AdamW keeps the local second-moment target independent of the proposal, and GPAS chooses the proposal in that fixed geometry. Applying the allocation after each objective update also makes the method compatible with a curriculum.

AI use statement

We used generative AI tools for language editing and manuscript organization. The authors reviewed every AI-assisted revision and take responsibility for the final text, claims, proofs, code, and experimental results.

Ethics statement

The study evaluates optimization methods for existing language models with established reasoning and instruction-following datasets and automated verifiers. Model and dataset biases can affect the reported capability scores. We record data provenance, invalid outputs, and verifier failures to support careful interpretation of the results.

Reproducibility statement

Section 5 specifies the model family, task setup, update budget, evaluation points, and evaluation protocol. The appendix gives the sampling derivations and implementation details. The repository includes the controlled experiment code, all generated CSV files, and figure-generation code. This draft limits its numerical claims to the controlled studies because its MOPD checkpoints, gradient bank, and logs are outside the current artifact. Every large-model number added to the paper will be paired with a data manifest, model and tokenizer revisions, optimizer settings, task-sampling log, saved model state, and per-prompt evaluation output.

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A Practical Training Recipe

The recipe uses one task-homogeneous batch and one optimizer step after each rollout. Task selection occurs before generation, so a step calls one teacher or reward service.

1. Fix the teacher-loss objective. Choose λ for the weighted teacher-loss objective and record it before choosing a sampler. Benchmark evaluation measures the resulting capability profile separately.

2. Define one task gradient. Fix the token and response aggregation rule inside each task. Our MOPD setup averages tokens inside each response and then averages responses. It retains the existing fixed-token batch budget and records the realized response count, prompt count, valid-token count, and GPU time for each task.

3. Initialize the sampler. Run a short round-robin warm start. For each task, store a moving average of raw gradient norm and preconditioned gradient scale. Cost-GPAS also stores a moving average of complete step time. The four-task protocol uses eight warm-start steps and $\gamma = 0.95$.

4. Keep the raw gradient and weight separate. After sampling task I, compute its raw task gradient G_I and the factor $w = \lambda_I / q_I$. Moment-consistent AdamW uses wG_I for the first moment and wG_I^2 for the second moment. Passing only wG_I to an unchanged AdamW implementation produces the standard $w^2 G_I^2$ second moment. The optimizer interface therefore accepts both G_I and w.

5. Update and save the sampler. GPAS uses $q_i \propto \lambda_i \widehat{s}_i$; Cost-GPAS uses $q_i \propto \lambda_i \widehat{s}_i / \sqrt{\widehat{c}_i}$. For raw proposal p, apply $q_i = (1 - M q_{\min}) p_i + q_{\min}$ before drawing a task. Save scale and cost averages with model and optimizer state. Log I, p_I, q_I, w, valid tokens, response count, complete step time, raw norm, preconditioned norm, corrected update norm, scale age, and clipping status.

B Derivations and Evaluation Details

B.1 Moment targets

For Lemma 1, conditioning on the state before the draw gives

$$\mathbb{E}[w_I G_I] = \sum_i q_i \frac{\lambda_i}{q_i} \mathbb{E}[G_i] = \sum_i \lambda_i \mathbb{E}[G_i]. \tag{17}$$

The same calculation with an element-wise square gives

$$\mathbb{E}[w_I G_I^2] = \sum_i \lambda_i S_i. \tag{18}$$

Standard AdamW squares the complete corrected gradient, so

$$\mathbb{E}[w_I^2 G_I^2] = \sum_i \frac{\lambda_i^2}{q_i} S_i. \tag{19}$$

These are the observations entering the exponential average in Proposition 1.

B.2 GPAS DERIVATION AND THE DESCENT CONNECTION

Let A be the fixed diagonal scale with entries $1/(\sqrt{\bar{v}} + \epsilon)$, let $h_i = \mathbb{E}[G_i]$, and let $h = \sum_i \lambda_i h_i$. The centered scaled error of one corrected draw is

$$\mathbb{E}\|A(\widehat{g} - h)\|_2^2 = \sum_i \frac{\lambda_i^2 s_i^2}{q_i} - \|Ah\|_2^2. \tag{20}$$

The final term is fixed before sampling. Cauchy–Schwarz gives

$$\sum_i \frac{\lambda_i^2 s_i^2}{q_i} \geq \left(\sum_i \lambda_i s_i \right)^2, \tag{21}$$

with equality when q_i is proportional to $\lambda_i s_i$.

For the local optimization connection, consider the preconditioned update $\theta^+ = \theta - \eta A \widehat{g}$ and an L-smooth objective. The descent lemma and unbiasedness give

$$\mathbb{E}[F(\theta^+)] \leq F(\theta) - \eta \langle h, Ah \rangle + \frac{L\eta^2}{2} \sum_i \frac{\lambda_i^2 s_i^2}{q_i}. \tag{22}$$

At a fixed state and learning rate, the last term contains all dependence on the proposal. AdamW momentum, changing preconditioners, and weight decay make this a local comparison rather than a trajectory theorem.

B.3 COST CONNECTION

Write $\bar{c}(q) = \sum_i q_i c_i$. A time budget B gives approximately $T = B/\bar{c}(q)$ steps. A standard smooth non-convex stochastic bound has the form

$$\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}\|\nabla F(\theta_t)\|^2 \leq \frac{C_1}{\eta T} + C_2 \eta V(q), \tag{23}$$

for constants independent of q under a fixed, bounded preconditioner. Tuning η in this bound leaves a q-dependent factor proportional to $\sqrt{\bar{c}(q) V(q)/B}$. This motivates the product $J(q)$ as a compute-efficiency proxy.

A second Cauchy–Schwarz calculation gives

$$J(q) = \left(\sum_i q_i c_i \right) \left(\sum_i \frac{\lambda_i^2 s_i^2}{q_i} \right) \geq \left(\sum_i \lambda_i s_i \sqrt{c_i} \right)^2. \tag{24}$$

Equality holds at $q_i \propto \lambda_i s_i / \sqrt{c_i}$.

B.4 SAMPLED-TOKEN MOPD GRADIENT

Fix a student-generated prefix z. Write $p_v = \pi_\theta(v \mid z)$ and $\tau_v = T_i(v \mid z)$. The reverse-KL loss gradient at that prefix is

$$g_i(z) = \sum_v p_v \log \frac{p_v}{\tau_v} \nabla_\theta \log p_v. \tag{25}$$

For $V \sim p$, the sampled-token gradient

$$G_i(z, V) = \mathrm{sg}\left[\log \frac{p_V}{\tau_V}\right] \nabla_\theta \log p_V \tag{26}$$

has conditional mean $g_i(z)$. The implementation averages these token gradients inside each response and then averages responses to form G_i.

B.5 Frozen-gradient evaluation

At one checkpoint, the gradient bank estimates each task mean h_i, raw second moment, preconditioned second moment s_i^2, and runtime c_i. Proposal scales are estimated on one fold and evaluated on the other. Reversing the folds and averaging uses every batch while preserving held-out evaluation. The primary local quantities are $V(q)$ and $J(q)$ because they avoid the upward bias of a sample mean norm in centered MSE. A split estimate of Equation 20 is reported as a secondary check.

To evaluate update prediction, each held-out G_i is inserted into the saved AdamW state while every other input is held fixed. The task-dependent update is $\Delta\theta(G_i) - \Delta\theta(0)$, which removes weight decay and state-only terms. Pearson and Spearman correlations are computed within tasks and on the pooled bank. Pairwise cosine similarity uses task mean gradients under one shared random projection.

For state replay, a task sequence is drawn from each proposal. Standard AdamW accumulates $w^2 G_i^2$, and moment-consistent AdamW accumulates $w G_i^2$. Population-target error is computed directly from taskwise second moments; finite-memory drift compares the replayed state with the state from $q = \lambda$, normalized by the baseline state norm. A fixed sampled set of parameter coordinates keeps storage bounded.

B.6 Scale-estimation error and online feedback

For positive costs and a proposal before flooring, define $\widetilde{q}_i = q_i c_i / (\sum_j q_j c_j)$. Let q^* use current all-task scales and let $\widehat{q}$ use online estimates. Direct substitution gives

$$\frac{J(\widehat{q})}{J(q^*)} = 1 + D_{\chi^2}(\widetilde{q}^* \,\|\, \widetilde{\widehat{q}}). \tag{27}$$

This identity diagnoses scale-estimation error at one fixed state. Online training moves the reference scales, and the probability floor adds a separate allocation cost. We log reference age and report the floor contribution separately.

B.7 Strict curriculum decomposition

An online curriculum can generate a different objective sequence after the training path changes. Native D^3 first records its sequence $\lambda_1, \ldots, \lambda_T$ for each seed. The paired D^3 target + GPAS run replays that sequence and changes only q_t. Both runs share initialization, per-task data order, and evaluation prompts.

D^3-Proposal-IC serves a different comparison. It keeps $\lambda_i = 1/4$, forms q_t from the published loss-gap and descent-velocity signal, and applies $\lambda_i / q_{t,i}$. Its comparison with GPAS measures two proposal signals under one weighted teacher-loss objective.

B.8 Sampler state, clipping, and resume

Each checkpoint stores the scale average, runtime average, observation count, scale age, probability floor, and current proposal for every task. The optimizer checkpoint stores the usual AdamW state and a flag identifying the second-moment rule. Resume tests compare the next proposal and optimizer state before and after serialization.

The theory concerns the raw teacher-loss gradient. Training applies one shared emergency clip before importance correction. This nonlinear map changes the mean relative to the original loss gradient when clipping fires, while the λ/q correction stays unbiased for the clipped update field. Every method reports clip rate by task, multiplier quantiles, corrected-update norm quantiles, and the fraction of steps affected.